1.39 2D Matrices: Shears

Worked Solutions

1 a)

Shear, with the x-axis fixed, with $(0, 1)$ mapped to $(2, 1)$.

1 b)

Shear, with the y-axis fixed, with $(1, 0)$ mapped to $(1, 3)$.

1 c)

Shear, with the x-axis fixed, with $(0, 1)$ mapped to $(-4, 1)$.

1 d)

Shear, with the y-axis fixed, with $(1, 0)$ mapped to $(1, -\frac{3}{2})$.

2)

This is a shear, with the x-axis fixed, with $(0, 1)$ mapped to $(6, 1)$, so area is preserved. Hence the area of the shape's image is 4 units^2.

3)

$$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2 \\ 4 \end{pmatrix} = \begin{pmatrix} 18 \\ 4 \end{pmatrix} \Rightarrow \begin{pmatrix} -2 + 4k \\ 4 \end{pmatrix} = \begin{pmatrix} 18 \\ 4 \end{pmatrix}$$

$-2 + 4k = 18 \Rightarrow k = 5$.

4)

$$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \begin{pmatrix} 22 \\ -6 \end{pmatrix} = \begin{pmatrix} 22 \\ -94 \end{pmatrix} \Rightarrow \begin{pmatrix} 22 \\ 22k - 6 \end{pmatrix} = \begin{pmatrix} 22 \\ -94 \end{pmatrix}$$

$22k - 6 = -94 \Rightarrow k = -4$.

5)

Triangle X has area

$$\frac{1}{2} \times 7\sqrt{2} \times \sqrt{2} = 7 \text{ units}^2$$

$$\begin{pmatrix} 1 & 8\sqrt{2} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & -3\sqrt{2} & 4\sqrt{2} \\ \sqrt{2} & 0 & 0 \end{pmatrix} = \begin{pmatrix} 15 & -3\sqrt{2} & 4\sqrt{2} \\ \sqrt{2} & 0 & 0 \end{pmatrix}$$

So X' has vertices at $(15, \sqrt{2})$, $(-3\sqrt{2}, 0)$ and $(4\sqrt{2}, 0)$.

Triangle X' has area

$$\frac{1}{2} \times 7\sqrt{2} \times \sqrt{2} = 7 \text{ units}^2$$

6)

$$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \begin{pmatrix} 8 \\ 3\sqrt{3} \end{pmatrix} = \begin{pmatrix} 8 \\ 5\sqrt{3} \end{pmatrix} \Rightarrow \begin{pmatrix} 8 \\ 8k + 3\sqrt{3} \end{pmatrix} = \begin{pmatrix} 8 \\ 5\sqrt{3} \end{pmatrix}$$

$8k + 3\sqrt{3} = 5\sqrt{3} \Rightarrow 8k = 2\sqrt{3} \Rightarrow k = \frac{\sqrt{3}}{4}$.

$\therefore \mathbf{A} = \begin{pmatrix} 1 & 0 \\ \frac{\sqrt{3}}{4} & 1 \end{pmatrix}$.

7)

$$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 50 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} \Rightarrow \begin{pmatrix} 50 + 4k \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$$

$50 + 4k = 4 \Rightarrow 4k = -46 \Rightarrow k = -\frac{23}{2}$.

$\therefore \mathbf{A} = \begin{pmatrix} 1 & -\frac{23}{2} \\ 0 & 1 \end{pmatrix}$.

8)

$$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 45 \\ 9 \end{pmatrix} = \begin{pmatrix} 0 \\ p \end{pmatrix} \Rightarrow \begin{pmatrix} 45 + 9k \\ 9 \end{pmatrix} = \begin{pmatrix} 0 \\ p \end{pmatrix}$$

$45 + 9k = 0 \Rightarrow 9k = -45 \Rightarrow k = -5$

$p = 9$

$\therefore \mathbf{A} = \begin{pmatrix} 1 & -5 \\ 0 & 1 \end{pmatrix}$.

9)

As $(5, 0)$ is mapped to itself, the shear must fix the x-axis.

$$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 20 \end{pmatrix} = \begin{pmatrix} 11 \\ 20 \end{pmatrix} \Rightarrow \begin{pmatrix} 3 + 20k \\ 20 \end{pmatrix} = \begin{pmatrix} 11 \\ 20 \end{pmatrix}$$

$3 + 20k = 11 \Rightarrow 20k = 8 \Rightarrow k = \frac{2}{5}$.

$\therefore \mathbf{A} = \begin{pmatrix} 1 & \frac{2}{5} \\ 0 & 1 \end{pmatrix}$.

10 a)

As $(-12, 8)$ is mapped to $(-12, 17)$, the shear must fix the y-axis.

$$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \begin{pmatrix} -12 \\ 8 \end{pmatrix} = \begin{pmatrix} -12 \\ 17 \end{pmatrix} \Rightarrow \begin{pmatrix} -12 \\ -12k + 8 \end{pmatrix} = \begin{pmatrix} -12 \\ 17 \end{pmatrix}$$

$-12k + 8 = 17 \Rightarrow -12k = 9 \Rightarrow k = -\frac{3}{4}$.

$\therefore \mathbf{A} = \begin{pmatrix} 1 & 0 \\ -\frac{3}{4} & 1 \end{pmatrix}$.

10 b)

As $(-12, 17)$ is mapped to $(-12, 8)$, the shear must fix the y-axis.

$$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \begin{pmatrix} -12 \\ 17 \end{pmatrix} = \begin{pmatrix} -12 \\ 8 \end{pmatrix} \Rightarrow \begin{pmatrix} -12 \\ -12k + 17 \end{pmatrix} = \begin{pmatrix} -12 \\ 8 \end{pmatrix}$$

$-12k + 17 = 8 \Rightarrow -12k = -9 \Rightarrow k = \frac{3}{4}$.

$\therefore \mathbf{B} = \begin{pmatrix} 1 & 0 \\ \frac{3}{4} & 1 \end{pmatrix}$.

10 c)

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ -\frac{3}{4} & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \frac{3}{4} & 1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 0 \times \frac{3}{4} & 1 \times 0 + 0 \times 1 \\ -\frac{3}{4} \times 1 + 1 \times \frac{3}{4} & -\frac{3}{4} \times 0 + 1 \times 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}$$

11)

The transformation is a shear, and clearly it is with the y-axis fixed, with $(1, 0)$ mapped to $(1, k)$.

$$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 7 & 5 \\ 2 & 6 & 6 & 2 \end{pmatrix} = \begin{pmatrix} a & b & c & 5 \\ d & e & f & 8.25 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & 4 & 7 & 5 \\ 2k + 2 & 4k + 6 & 7k + 6 & 5k + 2 \end{pmatrix} = \begin{pmatrix} a & b & c & 5 \\ d & e & f & 8.25 \end{pmatrix}$$

$\therefore a = 2, b = 4, c = 7$

$5k + 2 = 8.25 \Rightarrow k = 1.25$

$\therefore \mathbf{M} = \begin{pmatrix} 1 & 0 \\ 1.25 & 1 \end{pmatrix}$

$A'(2, 4.5)$, $B'(4, 11)$, $C'(7, 14.75)$.

12)

The transformation is a shear, and clearly it is with the x-axis fixed, with $(0, 1)$ mapped to $(k, 1)$.

$$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & -5 & -5 & -1 \\ 3 & 6 & 9 & 9 \end{pmatrix} = \begin{pmatrix} a & b & c & 5 \\ d & e & f & 9 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -1 + 3k & -5 + 6k & -5 + 9k & -1 + 9k \\ 3 & 6 & 9 & 9 \end{pmatrix} = \begin{pmatrix} a & b & c & 5 \\ d & e & f & 9 \end{pmatrix}$$

$\therefore d = 3, e = 6, f = 9$

$-1 + 9k = 5 \Rightarrow k = \frac{2}{3}$

$\therefore \mathbf{M} = \begin{pmatrix} 1 & \frac{2}{3} \\ 0 & 1 \end{pmatrix}$

$A'(1, 3)$, $B'(-1, 6)$, $C'(1, 9)$.

13 a)

Shear, with the y-axis fixed, with $(1, 0)$ mapped to $(1, k)$.

13 b)

$\det \mathbf{A} = 1 \times 1 - 0 \times k = 1$.

13 c)

As $\det \mathbf{A} = 1$, both area and orientation are preserved by the transformation.